CRICKETLENS / IT462 PROJECT

Stage 2 — Consolidated Exploratory Data Analysis

Sessions 01–03 | Men’s IPL 2008–2025 | Outcomes, distributions, statistical comparisons and exceptional-match handling

Executive summary

CricketLens is an EDA-first men’s IPL analytics project with two planned probabilistic prediction models. This report records verified descriptive work and fixes a genuine analytical error: regulation ties decided by a Super Over were previously excluded from team win statistics. The corrected pipeline preserves regulation results and credits the final match winner separately.

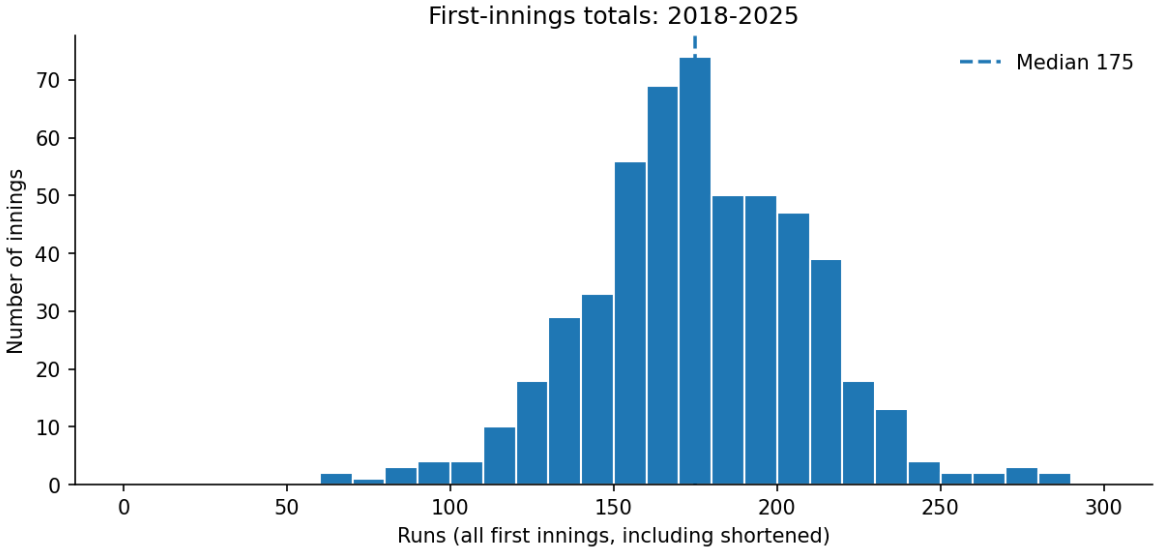
Analysis period	Coverage and principal evidence
Historical development	1,169 IPL matches (2008–2025); IPL 2026 held out for ML testing
Regulation outcomes	1,146 regulation wins; 15 Super Over-decided ties; 8 no-results
D/L method annotation	22 pre-2026 matches; retained and analyzed separately, not deleted
First innings, 2018–2025	533; mean 175.80, median 175 runs
Scoring phase comparison	Death minus Powerplay +2.26 runs per six legal balls in paired innings
Outcome accounting correction	Delhi Capitals 26/55 decided matches won (2022–2025), not 25/54

Status and academic integrity

Stage 2 numerical EDA is complete and ready for a professor’s review; the website and both final ML models are not yet built. Results are observational associations, not causal claims or final 2026 forecast evaluations. Source: Cricsheet men’s IPL JSON; project analysis/code credited as documented.

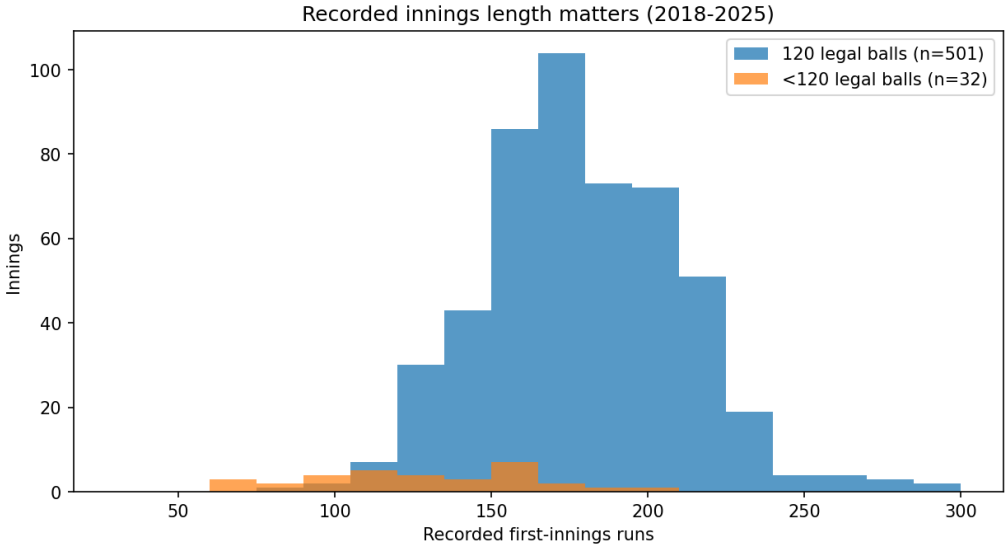
How scoring patterns vary across IPL innings

Between 2018 and 2025, 533 recorded first innings averaged 175.80 runs, with median 175. Their sample SD was approximately 33.93; Q1 155, Q3 199, and IQR 44. Eleven scores fall outside the Tukey fences (<89 or >265). These records are retained unless validated as data errors.



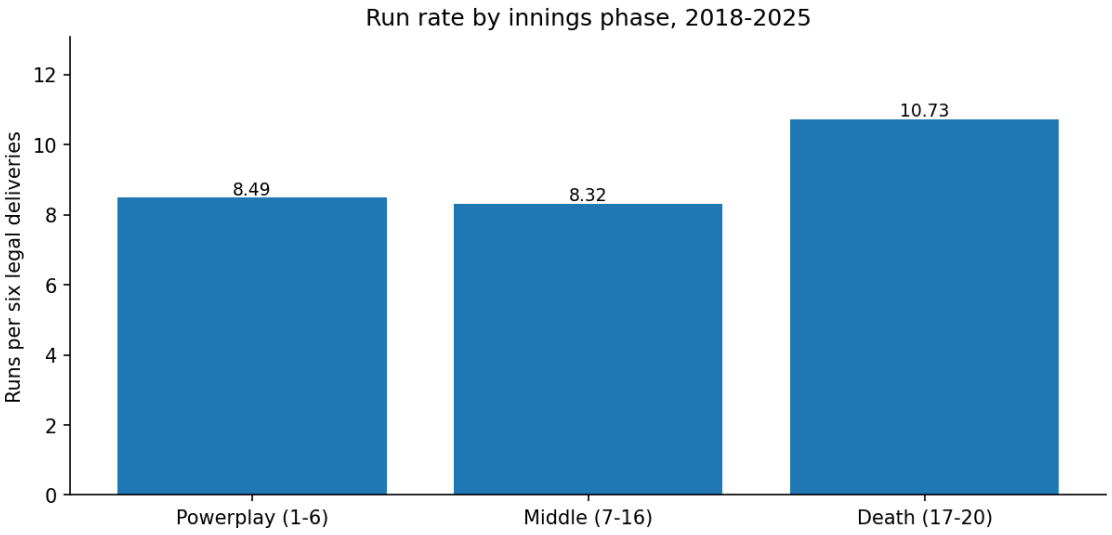
Innings duration changes the comparison

501 full 120-legal-ball first innings averaged 178.88 runs; 32 shorter ones averaged 127.50. An innings may end early due to all-out, match conditions or abandonment, so a low score is not automatically a weak 20-over batting performance.



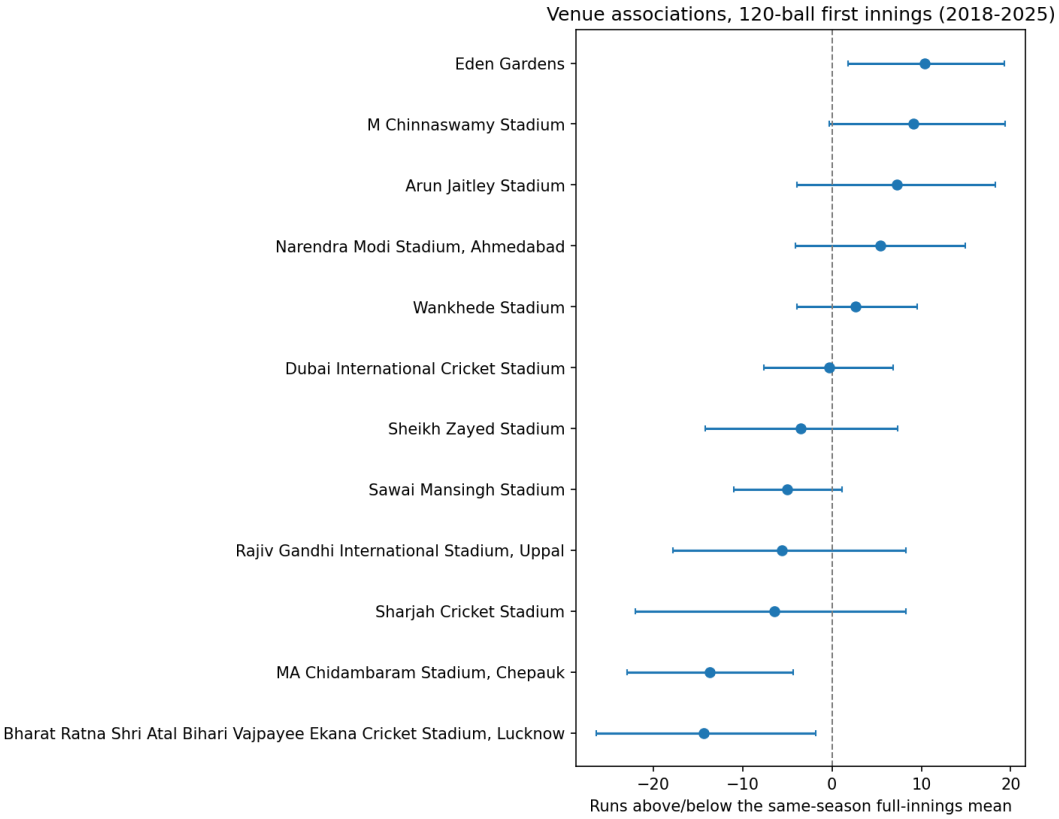
Patterns supported by statistical comparisons

The aggregate observed run rate was 8.49 (Powerplay), 8.32 (middle) and 10.73 (death), measured as runs per six legal deliveries. In a paired analysis of 937 innings with sufficient balls in both relevant phases, death overs averaged +2.26 versus Powerplay (approximate 95% match-clustered bootstrap interval +2.02 to +2.51). This is conditional on reaching the death phase.



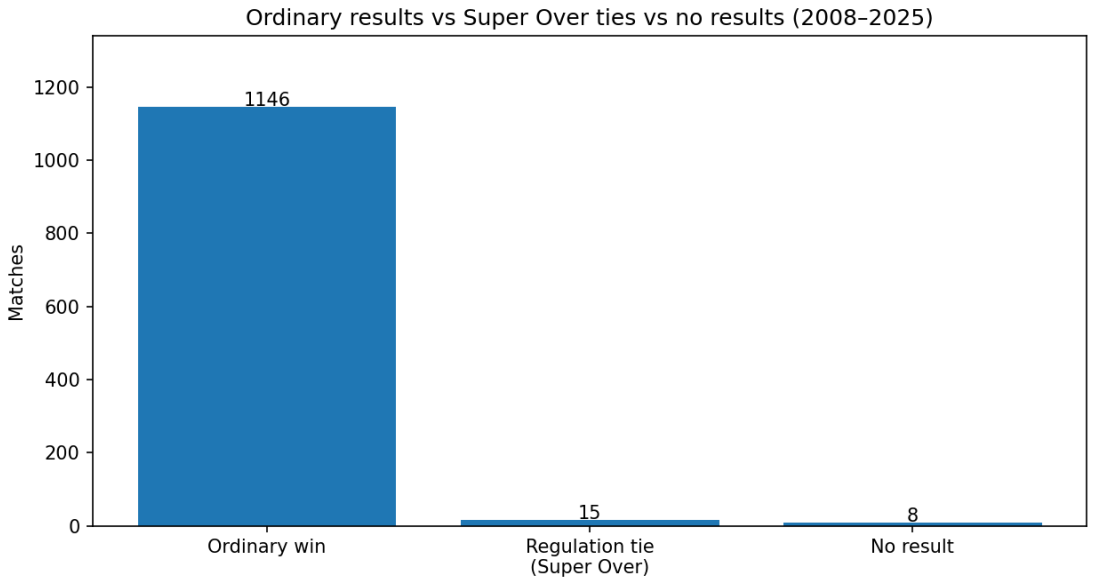
Venue and seasonal confounding

Among full 120-ball first innings, Eden Gardens scored +10.34 runs relative to the same-season average (n=37), versus Chepauk at -13.71 (n=38). Venue, team, pitch, weather and year are not randomized: the contrasts are associations, not causal effects. A simple line fitted to yearly means shows about +1.71 runs per season over 2008–2025, also descriptive.



Super Overs and no-results — now represented correctly

A tied match can still have a final winner. Cricsheet provides an eliminator field identifying the Super Over winner. We added regulation_result, super_over_winner, final_winner, decided_by, and super_over_innings, keeping the original winner for provenance.

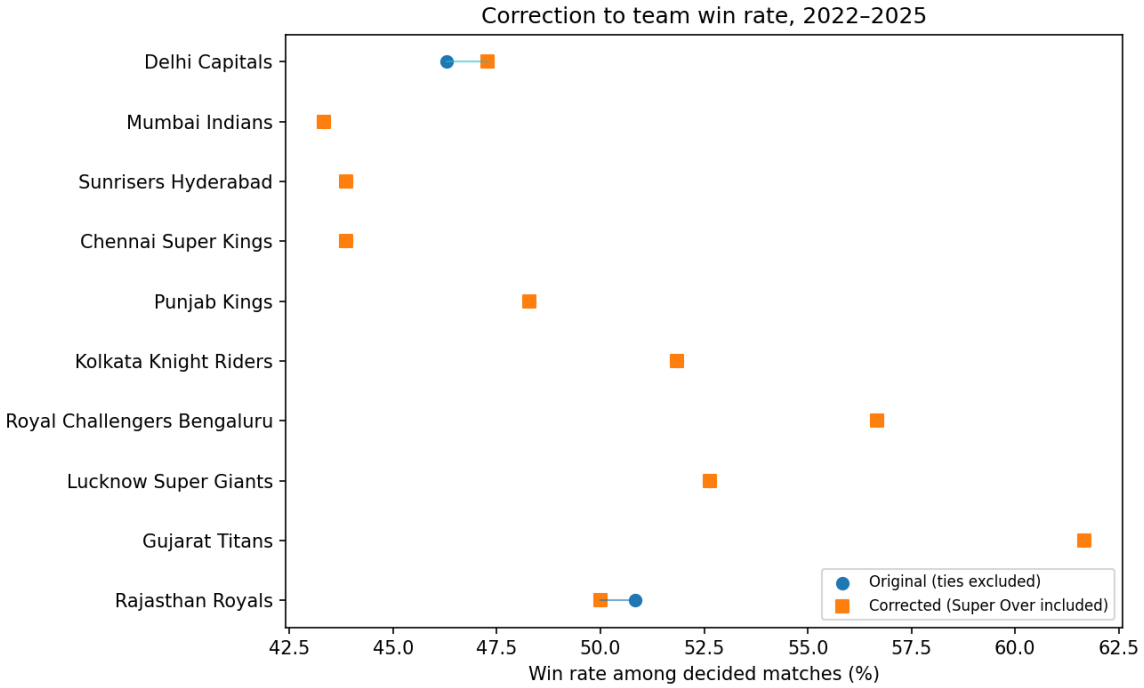


2008–2025 record type	Count	Handling
Ordinary regulation win	1,146	Final winner is the match winner
Regulation tie; Super Over decides	15	Preserve tie; assign Super Over winner
No result	8	Keep match and any recorded events; no invented winner
D/L-marked match	22	Separate outcome-method flag (may overlap a win)

The complete 2008–2026 archive was also checked mechanically: 16 Super Over-decided ties (34 separate Super Over innings), 9 no-results and 23 D/L-marked results. These all-archive counts are input-quality checks, not development analysis of the 2026 test set.

Why the Super Over correction changes conclusions

Previously the team-win table discarded every ordinary tie, even when the match had a documented winner from a Super Over. In the 2022–2025 team window, correcting the final-match outcome adds a victory for Delhi Capitals, changing its win rate from 25/54 = 46.30% to 26/55 = 47.27%. Rajasthan Royals' denominator increases to 60 (30 wins), lowering its calculated rate from 50.85% to 50.00%.



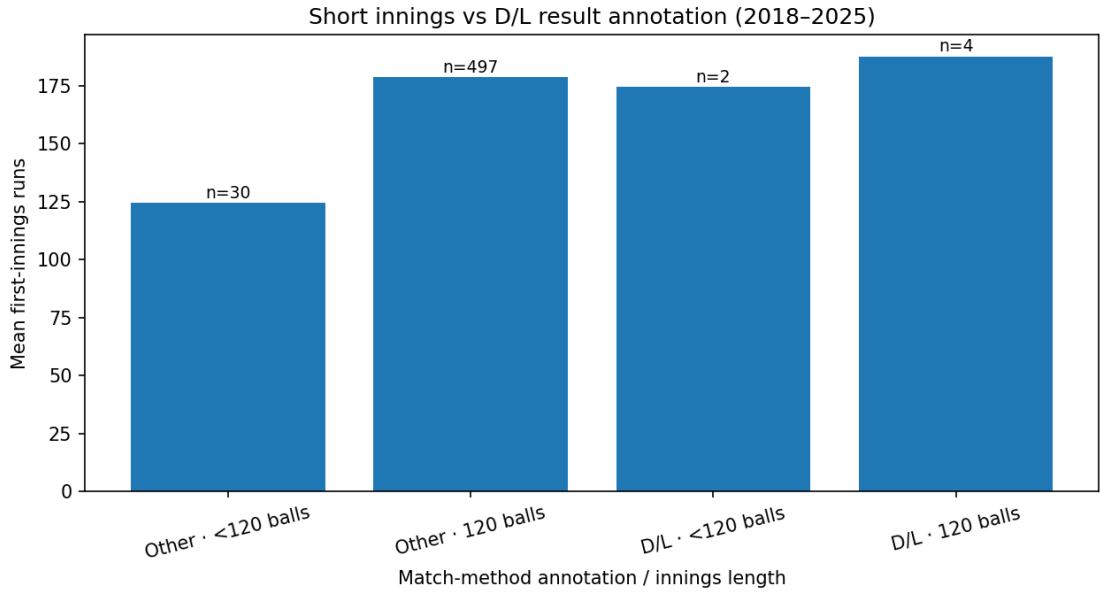
Toss association — corrected denominator

From 2018–2025 the team winning the toss also won 62/130 (47.69%) of matches where it chose to bat, compared with 210/398 (52.76%) choosing to field. Fisher's two-sided test gives p≈0.363. This cannot establish a causal advantage from the toss decision.

Erratum: The earlier Session 02 PDF reported 60/127 and 209/393 because Super Over winners were omitted. Use this consolidated report and the regenerated Session 02 CSVs instead.

Rain-adjusted match results are not the same as short innings

Cricsheet marks 22 development-period matches with the result method D/L. This is a match-level annotation: it does not reveal which innings was interrupted or at what over. It would be incorrect to simply multiply shortened innings scores into imaginary 20-over totals, or to assume every short innings was affected by rain.



First innings, 2018–2025	120 legal balls	Fewer than 120
No D/L result annotation	497	30
D/L result annotation	4	2

The recent sample contains only six D/L-marked first innings, too few for a reliable subgroup effect estimate. No-result records retain available deliveries, but contribute no winner to win-rate denominators. In the 1,169 pre-2026 match records, 51 city fields are missing (33 Dubai and 18 Sharjah); venue names are preserved and the missing city is not invented.

MCAR, MAR and MNAR mechanisms are considered but not proven. Observed association of missing city with venue makes a uniform missingness mechanism implausible, but the mechanism cannot be uniquely identified.

EDA methods demonstrated

Syllabus topic	Work performed
Screening / transformation	Parsed 1,243 match records; audited legal deliveries, venue aliases, player IDs and exceptions
Measurement scales / sampling	Nominal, ordinal, discrete ratio and time index documented; era and selection bias discussed
Missingness / outliers	Missing-city mapping; structural missing winner/method; Tukey fences and trimmed mean
Descriptive statistics	Mean, median, mode, SD, IQR, variance, MAD, CV, five-number summary
Visualization / associations	Histograms, boxplots, scatters, phase plots, cross-tabulations, venue-adjusted comparisons
Correlation and fitting	Pearson/Spearman, descriptive multiple linear regression, VIF, goodness of fit
Advanced EDA / storytelling	Season trend, paired-phase uncertainty, match exception corrections, dashboard figure selection

Dashboard chart plan

Tab 1: innings distribution, annual score trends and run rates. Tab 2: corrected team wins and venue trends. Tab 3: batter milestones, volume and consistency. Tab 4: boxplots, bivariate correlations, missingness, exceptions and season-centered comparisons. Tab 5: descriptive wicket-rate context alongside the future trained models. Tab 6: calibration and recorded-vs-actual summaries, to be built after model training.

What remains for Stage 3

Define proper next-over target censoring when an innings ends early, exclude/stratify D/L-affected matches appropriately, construct leakage-safe player form, train baselines and calibrated classifiers, validate on 2025, and score the held-out 2026 data once after model selection. Our two approved models remain: batter 30+/50+ probabilities and next-over wicket probability.

Reproducibility and limitations

Repository modules: cricketlens.pipeline, stage02_eda, stage02_deep_eda, stage02_final_eda. Outputs: reports/stage_02/session_01–03 and reports/figures/stage_02. Run pytest before merging the feature branch. The old model baseline has no final test result; no online tracking or retraining was implemented here.

Source attribution: Cricsheet men’s IPL JSON; academic work should acknowledge any AI assistance according to IT462 policy. The EDA is internally complete for code review; the professor has not yet reviewed or approved its conclusions.